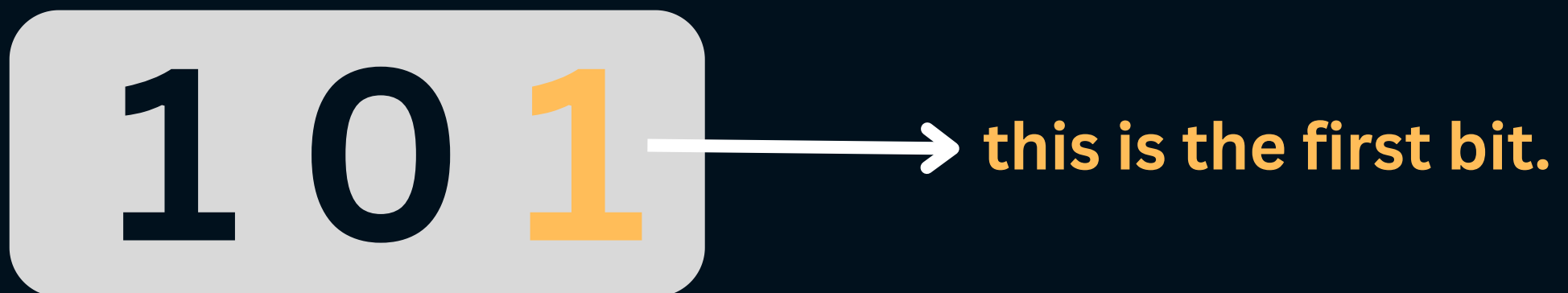


Clear ith Bit

Give a Number **N**, the task is to clear the **i-th** bit of the Number **N**.

Examples: **N=5, i=1**

- **5** is represented as **101** in binary representation.



1 0 0



this is the result ,
after clear ith bit.

- write **bit_mask = (1<<(k-1))**
- k - 1 adjusts the 1-indexed position to a 0-indexed position.



```
1
2 // C++ program to clear K-th bit of a number N
3
4 #include <bits/stdc++.h>
5 using namespace std;
6
7 int clearBit(int n, int k)
8 {
9     int bit_mask = ~(1<<(k-1));
10    int result = n & bit_mask;
11    return result;
12 }
13
14 int main()
15 {
16     int n = 5, k = 1;
17
18     cout<<clearBit(n, k)<<endl;
19
20     return 0;
21 }
22
```

code in c++



```
1  function clearBit(n, k) {  
2      let bitMask = ~(1 << (k - 1));  
3      let result = n & bitMask;  
4      return result;  
5  }  
6  
7  function main() {  
8      let n = 5;  
9      let k = 1;  
10  
11      console.log(clearBit(n, k));  
12  }  
13  
14  main();
```

code in javascript



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